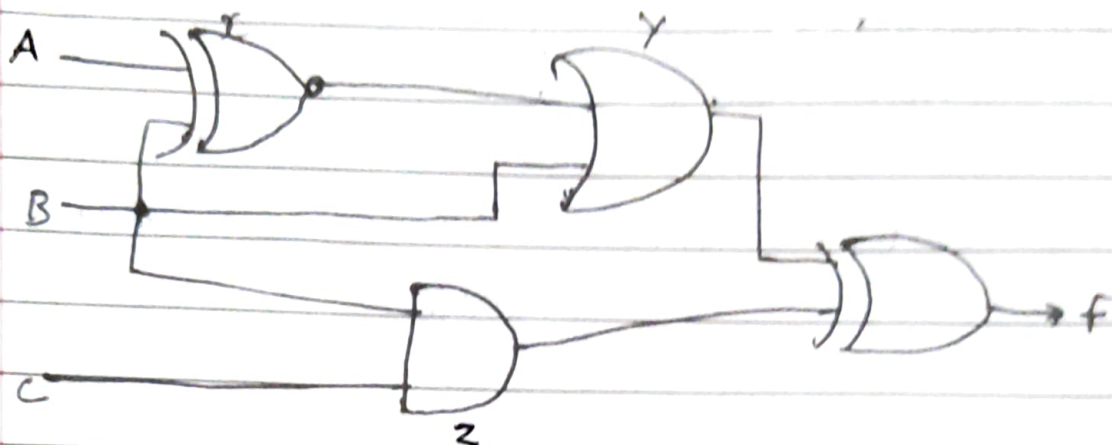


Q)1. Derive the expression 'f'

a)



$$x = (A \oplus B)$$

$$= \underline{AB + \bar{A}\bar{B}}$$

$$y = x + B$$

$$= (AB + \bar{A}\bar{B}) + B$$

$$y = B + \bar{A}\bar{B} = \underline{B + \bar{A}}$$

$$\therefore f = \underline{\bar{A} + BC}$$

$$z = BC$$

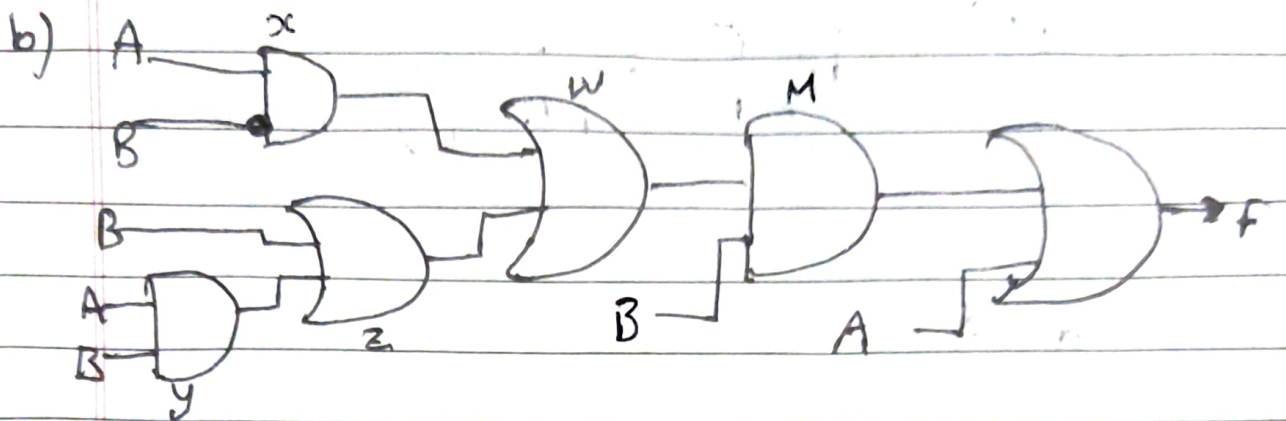
$$f = y + z$$

$$= (\bar{A} + B) + BC$$

$$= (\bar{A} + B)\bar{B}C + (\bar{A} + B)BC$$

$$= \underline{\bar{A} + BC}$$

$$\therefore P \oplus Q = P\bar{Q} + \bar{P}Q$$



$$x = A\bar{B}$$

$$y = AB$$

$$z = B + AB$$

$$z = B$$

$$w = x + z$$

$$w = A\bar{B} + B \quad (x + yz = (x + y)(x + z),)$$

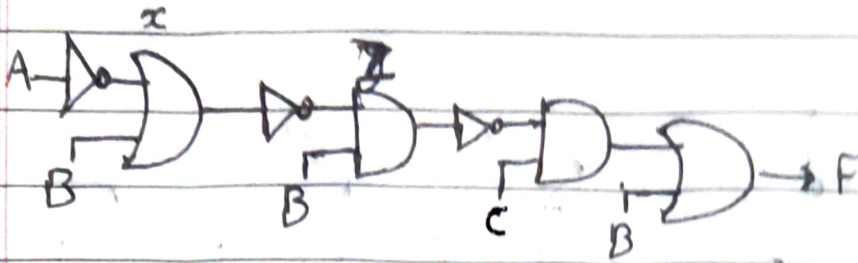
$$w = A + B$$

$$M = (A + B)B \\ = B$$

$$f = M + A \\ = A + B$$

$$\therefore f = \underline{\underline{A + B}}$$

c)



$$A = \bar{A}$$

$$X = \bar{A} + B$$

$$Y = \overline{(\bar{A} + B)} = A\bar{B}$$

$$Z = A\bar{B} \cdot B = 0$$

$$\bar{Z} = 1$$

$$1 \cdot C = C \quad f = C + B$$

$$\therefore f = \underline{\underline{B + C}}$$

Q/p Minimize the following Boolean Expressions

a) $\bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

Ans:- $f = \bar{A}\bar{B}\bar{D} + ABD + \bar{A}\bar{C}\bar{D}E + ABCE$

~~$f = D(\bar{A}\bar{B} + AB) + C\bar{D}E(\bar{A} + A)$~~

$f = \bar{A}\bar{B}\bar{D} + ABD + CE(\bar{A}\bar{D} + AB)$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}\bar{C}\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

Ans:- $\bar{A}C + \bar{A}\bar{C}\bar{D} = \bar{A}C$

$f = \bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D + A\bar{B}\bar{C}$

$\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} = \bar{B}\bar{C}$

$f = \bar{A}C + \bar{B}\bar{C} + A\bar{C}D + ABD$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

Ans:-

$$BC + \bar{A}BC = BC$$

$$f = \bar{A}\bar{C} + BC + A\bar{B} + AC\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$A\bar{B} + \bar{B}\bar{C}\bar{D} = \bar{B}(A + \bar{C}\bar{D})$$

$$f = \underline{\bar{A}\bar{C} + BC + A\bar{B}}$$

d) $\bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 x_2 \bar{x}_3 x_4$

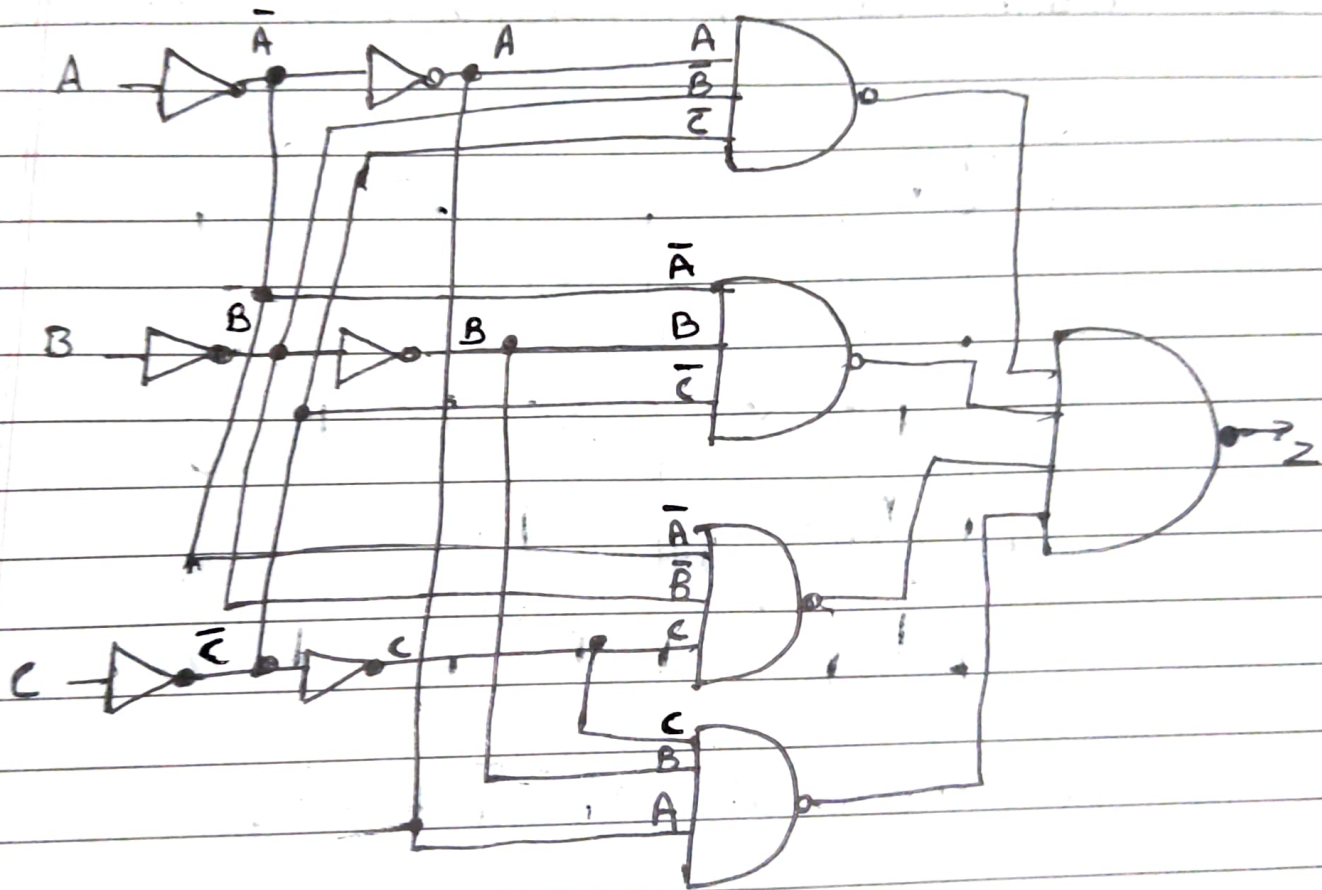
Ans:-

$$f = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_2$$

$$x_2 + \bar{x}_1 \bar{x}_2 x_4 = x_2 + \bar{x}_1 x_4$$

$$f = \underline{\underline{x_2 + \bar{x}_1 x_4 + x_3 x_4 + \bar{x}_1 \bar{x}_3 \bar{x}_4}}$$

Q)3 which single gate can represent the functionality



$$Z = [A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC]$$

$$\Rightarrow \underline{Z = A \oplus B \oplus C}$$

Q)4. Minimize the following B.E using K-map.

a) $f(a,b,c,d) = \sum m(4,6,8,10,11,12,15) + \sum d(3,5,7,9)$

Ans:

cd \ ab	00	01	11	10
00	0	1	X	2
01	1	X	X	6
11	1	3	1	14
10	1	X	1	10

$$f = \bar{a}b + a\bar{b} + b\bar{c}\bar{d} + acd$$

b) $f(a,b,c,d) = \prod M(0,2,4,6,7,9) \cdot \prod D(10,11)$
 $f = \sum m(1,3,5,8,12,13,14,15)$

Ans:

ab \ cd	00	01	11	10
00	0	1	1	0
01	0	1	0	0
11	1	1	1	1
10	1	0	X	X

$$f = ab + a\bar{d} + \bar{a}bd + \bar{a}\bar{c}d$$

$$f = \underline{ab + a\bar{d} + \bar{a}d(\bar{b} + \bar{c})}$$